

Challenge 39: Alteration of cavity field coherences

Steady-state cavity coherences $\langle n' | \hat{\rho}_{c,ss} | n \rangle$, with a squeezed-input bonus

Companion implementation: `challenges/39/answer.py`

1 Statement

A three-level atom with ground states $|b\rangle, |d\rangle$ and excited state $|e\rangle$ sits in a lossless cavity resonant with the bright transition $|b\rangle \leftrightarrow |e\rangle$. With $\hbar = 1$,

$$\hat{H} = \frac{g}{2} \left(|b\rangle\langle e| \hat{a}^\dagger + |e\rangle\langle b| \hat{a} \right), \quad \hat{J} = \sqrt{\gamma} |d\rangle\langle e|, \quad \dot{\hat{\rho}} = -i[\hat{H}, \hat{\rho}] + \hat{J} \hat{\rho} \hat{J}^\dagger - \frac{1}{2} \{ \hat{J}^\dagger \hat{J}, \hat{\rho} \}, \quad (1)$$

and $\hat{\rho}_0 = |b\rangle\langle b| \otimes |\alpha\rangle\langle\alpha|$. We want $\langle n' | \hat{\rho}_{c,ss} | n \rangle$.

Careful: $\mathcal{L}\hat{\rho} = 0$ does not determine the answer

The Liouvillian of Eq. (1) has a massively degenerate kernel: every $|d\rangle\langle d| \otimes \hat{\sigma}$ with arbitrary cavity state $\hat{\sigma}$ is stationary, and so is $|b, 0\rangle\langle b, 0|$. Setting the left-hand side to zero therefore gives a family of solutions, and the physical steady state is selected by the *initial condition*. The clean way to impose “LHS = 0” is to integrate Eq. (1) over all time,

$$\int_0^\infty dt \dot{\hat{\rho}} = \hat{\rho}(\infty) - \hat{\rho}(0), \quad (2)$$

which turns the equations of motion into an *algebraic* linear system for the time integrals $\hat{M}_{xy} = \int_0^\infty \hat{\rho}_{xy}(t) dt$. That is the derivation below. (In the same vein, `qutip.steadystate()` returns an arbitrary kernel element here and must not be used — the numerics in Sec. 4 propagate to long times instead.)

2 Derivation

2.1 Block equations of motion

Write $\hat{\rho} = \sum_{x,y \in \{b,e,d\}} |x\rangle\langle y| \otimes \hat{\rho}_{xy}$, where each $\hat{\rho}_{xy}$ is an operator on the cavity. Substituting into Eq. (1) and collecting blocks,

$$\dot{\hat{\rho}}_{bb} = -\frac{ig}{2} \left(\hat{a}^\dagger \hat{\rho}_{eb} - \hat{\rho}_{be} \hat{a} \right), \quad (3)$$

$$\dot{\hat{\rho}}_{be} = -\frac{ig}{2} \left(\hat{a}^\dagger \hat{\rho}_{ee} - \hat{\rho}_{bb} \hat{a}^\dagger \right) - \frac{\gamma}{2} \hat{\rho}_{be}, \quad (4)$$

$$\dot{\hat{\rho}}_{eb} = -\frac{ig}{2} \left(\hat{a} \hat{\rho}_{bb} - \hat{\rho}_{ee} \hat{a} \right) - \frac{\gamma}{2} \hat{\rho}_{eb}, \quad (5)$$

$$\dot{\hat{\rho}}_{ee} = -\frac{ig}{2} \left(\hat{a} \hat{\rho}_{be} - \hat{\rho}_{eb} \hat{a}^\dagger \right) - \gamma \hat{\rho}_{ee}, \quad (6)$$

$$\dot{\hat{\rho}}_{dd} = \gamma \hat{\rho}_{ee}. \quad (7)$$

The blocks $\hat{\rho}_{bd}, \hat{\rho}_{ed}$ and their adjoints start at zero and are never sourced ($\hat{J}$ produces only $|d\rangle\langle d|$ terms, and $\hat{H}$ does not touch $|d\rangle$), so they stay zero: **no atomic coherence with the dark state is ever generated**, and the cavity state factorises as $\hat{\rho}_c = \hat{\rho}_{bb} + \hat{\rho}_{ee} + \hat{\rho}_{dd}$.

2.2 Integrating over all time

As $t \rightarrow \infty$ every excitation ends in $|d\rangle$ except the piece that started in the dark corner $|b, 0\rangle$ (for which $\hat{a}|b, 0\rangle = 0$, so it does not evolve). Hence

$$\hat{\rho}_{bb}(\infty) = \rho_{00}^0 |0\rangle\langle 0|, \quad \hat{\rho}_{be}(\infty) = \hat{\rho}_{ee}(\infty) = 0, \quad \hat{\rho}_c^{\text{ss}} = \rho_{00}^0 |0\rangle\langle 0| + \hat{\rho}_{dd}(\infty), \quad (8)$$

with $\rho_{n'n}^0 = \langle n' | \hat{\rho}_c(0) | n \rangle$. Integrating Eqs. (3)–(6) using (2) and (8) gives four algebraic equations for $\hat{M}_{bb}, \hat{M}_{be}, \hat{M}_{eb}, \hat{M}_{ee}$, and (7) gives the answer directly:

$$\hat{\rho}_{dd}(\infty) = \gamma \hat{M}_{ee}. \quad (9)$$

2.3 Solving, manifold by manifold

The operators $\hat{a}, \hat{a}^\dagger$ shift Fock indices by one, so the system closes on each pair (n, n') with $n, n' \geq 1$. Writing

$$\hat{M}_{bb} \supset u |n\rangle\langle n'|, \quad \hat{M}_{ee} \supset v |n-1\rangle\langle n'-1|, \quad \hat{M}_{be} \supset w |n\rangle\langle n'-1|, \quad \hat{M}_{eb} \supset x |n-1\rangle\langle n'|, \quad (10)$$

and abbreviating $\Omega_n = \frac{g}{2}\sqrt{n}$ (the matrix element $\langle e, n-1 | \hat{H} | b, n \rangle$), Eqs. (6), (4), (5), (3) become

$$\gamma v = -i(\Omega_n w - \Omega_{n'} x), \quad \frac{\gamma}{2} w = -i(\Omega_n v - \Omega_{n'} u), \quad (11)$$

$$\frac{\gamma}{2} x = -i(\Omega_n u - \Omega_{n'} v), \quad -\rho_{nn'}^0 = -i(\Omega_n x - \Omega_{n'} w). \quad (12)$$

Eliminating w, x from the two right-hand equations and inserting into the two left-hand ones gives, with $S = \Omega_n^2 + \Omega_{n'}^2$ and $P = \Omega_n \Omega_{n'}$,

$$v(\gamma^2 + 2S) = 4Pu, \quad \rho_{nn'}^0 = \frac{2}{\gamma}(Su - 2Pv) \implies \gamma v = \rho_{nn'}^0 \frac{2\gamma^2 P}{2(S^2 - 4P^2) + \gamma^2 S}. \quad (13)$$

With $\Omega_n^2 = g^2 n/4$ one has $S = \frac{g^2}{4}(n + n')$, $P = \frac{g^2}{4}\sqrt{nn'}$ and $S^2 - 4P^2 = \frac{g^4}{16}(n - n')^2$, so all powers of g cancel except one ratio:

$$\gamma v = \rho_{nn'}^0 K_{nn'}, \quad K_{nn'} = \frac{2\sqrt{nn'}}{\frac{g^2}{2\gamma^2}(n - n')^2 + n + n'}. \quad (14)$$

Note $K_{nn} = 1$: every $|b, n \geq 1\rangle$ ends up in $|d, n-1\rangle$ with probability one, as it must.

2.4 Result

Equation (14) says $\langle n-1 | \hat{\rho}_{dd}(\infty) | n'-1 \rangle = \rho_{nn'}^0 K_{nn'}$. Relabelling $n \rightarrow n' + 1$, $n' \rightarrow n + 1$ and restoring the dark corner of Eq. (8):

Result — any input state

$$\langle n' | \hat{\rho}_c^{\text{ss}} | n \rangle = \rho_{00}^0 \delta_{n0} \delta_{n'0} + \rho_{n'+1, n+1}^0 \frac{2\sqrt{(n+1)(n'+1)}}{\frac{g^2}{2\gamma^2}(n - n')^2 + n + n' + 2} \quad (15)$$

equivalently, in operator form with $\circ$ the elementwise (Hadamard) product in the Fock basis,

$$\hat{\rho}_c^{\text{ss}} = \rho_{00}^0 |0\rangle\langle 0| + F \circ \left(\hat{a} \hat{\rho}_c(0) \hat{a}^\dagger \right), \quad F_{n'n} = \frac{2}{\frac{g^2}{2\gamma^2}(n - n')^2 + n + n' + 2}. \quad (16)$$

For the coherent state of the problem, $\rho_{n'n}^0 = e^{-|\alpha|^2} \alpha^{n'} (\alpha^*)^n / \sqrt{n'! n!}$, and Eq. (15) becomes

$$\langle n' | \hat{\rho}_c^{\text{ss}} | n \rangle = e^{-|\alpha|^2} \left[\delta_{n0} \delta_{n'0} + \frac{2 |\alpha|^2 \alpha^{n'} (\alpha^*)^n}{\sqrt{n! n'}! \left[\frac{g^2}{2\gamma^2} (n - n')^2 + n + n' + 2 \right]} \right] \quad (17)$$

where $e^{-|\alpha|^2} \equiv \exp(-|\alpha|^2)$.

3 Physical content and checks

Trace. $\sum_n \langle n | \hat{\rho}_c^{\text{ss}} | n \rangle = e^{-|\alpha|^2} [1 + \sum_n |\alpha|^{2(n+1)} / (n+1)!] = e^{-|\alpha|^2} e^{|\alpha|^2} = 1$.

The diagonal is trivial and g, γ -independent. Setting $n = n'$ in Eq. (15) gives $\langle n | \hat{\rho}_c^{\text{ss}} | n \rangle = \rho_{n+1, n+1}^0 + \rho_{00}^0 \delta_{n0}$: exactly one photon is removed unless the cavity was already empty. All the physics of g and γ lives in the *coherences*.

The coherence-reduction factor. Dividing by $\sqrt{\rho_{nn}^{\text{ss}} \rho_{n'n'}^{\text{ss}}}$, the normalised coherence of the initially pure state is multiplied by

$$D_{n'n} = \frac{2\sqrt{(n+1)(n'+1)}}{\frac{g^2}{2\gamma^2} (n - n')^2 + n + n' + 2} \leq 1, \quad D_{nn} = 1, \quad (18)$$

by AM–GM. Two limits:

- $g/\gamma \rightarrow \infty$: $D \rightarrow 0$ for $n \neq n'$ — complete decoherence in the Fock basis. The emitted photon is a clock: the vacuum Rabi frequency $\Omega_n = \frac{g}{2}\sqrt{n}$ is resolved within the excited-state lifetime, so the emission time carries full which-path information about n .
- $g/\gamma \rightarrow 0$: $D \rightarrow 2\sqrt{(n+1)(n'+1)} / (n + n' + 2) < 1$ — decoherence survives even in the Purcell limit, now because the *rates* $\Gamma_n = 4\Omega_n^2/\gamma = g^2 n/\gamma$ differ with n : an exponential decay with an n -dependent time constant still leaks which-path information.

For example $D_{01} = 2\sqrt{2}/(3 + \frac{g^2}{2\gamma^2})$: 0.9412 at $g/\gamma = 0.1$, 0.5657 at $g/\gamma = 2$, 0.0534 at $g/\gamma = 10$.

4 QuTiP cross-check

`answer.py` propagates Eq. (1) with `qutip.mesolve` to $T = 60 / \min(\gamma/4, g^2/\gamma)$ (the slowest relaxation rate of the $n = 1$ manifold), traces out the atom, and compares with Eq. (15).

Two remarks make the comparison exact rather than approximate. First, a Fock cutoff at N is *not* an approximation here: $\hat{H}$ only converts $|b, n\rangle$ into $|e, n-1\rangle$, so the photon number never increases and $\text{span}\{|n\rangle : n < N\}$ is invariant. Second, the same truncated and renormalised amplitude vector is fed to both sides, so the column “formula (15)” below is a pure test of the physics, limited only by the integrator tolerances (`atol` = 10^{-12} , `rtol` = 10^{-10}); the “closed form” column additionally carries the $N \rightarrow \infty$ truncation error of Eq. (17), listed separately.

Structural checks at $g = 2, \gamma = 1, \alpha = 1.1, N = 24$: $\text{Tr } \hat{\rho}_c^{\text{ss}} = 1.0000000000000000$; $\max |\rho - \rho^\dagger| = 1.7 \times 10^{-18}$; $\max_n |\rho_{nn}^{\text{ss}} - (|c_{n+1}|^2 + |c_0|^2 \delta_{n0})| = 2.8 \times 10^{-17}$.

g	γ	α	N	$\max \rho^{(15)} - \rho^{\text{QuTiP}} $	$\max \rho^{(17)} - \rho^{\text{QuTiP}} $	truncation
1.0	1.0	1.0	20	1.36×10^{-11}	1.36×10^{-11}	1.15×10^{-11}
10.0	1.0	1.0	20	5.32×10^{-11}	5.32×10^{-11}	1.30×10^{-13}
0.2	1.0	1.2	22	2.50×10^{-13}	1.65×10^{-10}	1.65×10^{-10}
3.0	2.0	1.5	24	4.65×10^{-12}	1.01×10^{-10}	1.01×10^{-10}
2.0	0.7	$0.8 + 0.6i$	22	3.13×10^{-11}	3.13×10^{-11}	6.21×10^{-14}

Table 1: Coherent input. The parameter sweep spans deep weak coupling ($g/\gamma = 0.2$) to deep strong coupling ($g/\gamma = 10$) and a complex α . Generated by `answer.py`.

5 Bonus: squeezed input

Nothing in Sec. 2 used the coherent-state amplitudes — Eq. (15) is linear in $\hat{\rho}_c(0)$ and holds for *any* input, pure or mixed. For a squeezed vacuum $|\zeta\rangle = \hat{S}(re^{i\theta})|0\rangle$,

$$s_{2m} \equiv \langle 2m|\zeta\rangle = \frac{(-e^{i\theta} \tanh r)^m \sqrt{(2m)!}}{2^m m! \sqrt{\cosh r}}, \quad s_{2m+1} = 0, \quad (19)$$

so that

$$\langle n'|\hat{\rho}_c^{\text{ss}}|n\rangle = |s_0|^2 \delta_{n0} \delta_{n'0} + s_{n'+1} s_n^* \frac{2\sqrt{(n+1)(n'+1)}}{\frac{g^2}{2\gamma^2}(n-n')^2 + n + n' + 2} \quad (20)$$

with the striking consequence that **the steady state is supported on odd photon numbers plus the vacuum**: the squeezed vacuum contains only even n , and exactly one photon is always removed. The surviving vacuum weight is $|s_0|^2 = 1/\cosh r$. Table 2 confirms Eq. (20), and the last two rows of Table 3 confirm the general formula (15) for a squeezed *coherent* input $\hat{D}(\alpha)\hat{S}(r)|0\rangle$, for which both parities are populated.

g	γ	r	θ	N	$\max \rho^{(20)} - \rho^{\text{QuTiP}} $	$\text{Tr } \rho$	$P(\text{odd}) + P(0)$
1.0	1.0	0.6	0.0	24	2.07×10^{-11}	1.000000000000	1.0000000000
5.0	1.0	0.8	0.0	28	2.18×10^{-11}	1.000000000000	1.0000000000
0.3	1.0	0.5	0.0	24	2.12×10^{-12}	1.000000000000	1.0000000000
2.0	1.5	0.7	0.9	26	1.02×10^{-11}	1.000000000000	1.0000000000

Table 2: Squeezed-vacuum input. The last column verifies that all the population really does sit on odd n plus the vacuum.

g	γ	α	r	N	$\max \rho^{(15)} - \rho^{\text{QuTiP}} $
1.5	1.0	0.7	0.4	24	3.53×10^{-12}
3.0	1.0	$0.5 + 0.3i$	0.6	26	2.89×10^{-11}

Table 3: Squeezed coherent input $\hat{D}(\alpha)\hat{S}(r)|0\rangle$.

Worst disagreement over all eleven cases: 5.3×10^{-11} , consistent with the requested integrator tolerance.